

Federated E-values: Comparing Three Aggregation Strategies

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Code: <https://github.com/watilde/Harmonia>

Abstract

I built an open-source tool for federated E-value aggregation and tested three strategies on synthetic data. **One claim: sample-size weighting (FRI) converged from 2.015 (1k) to 2.149 (2.8m) as inter-site heterogeneity collapsed from 9.7% to 0.16%, with strategy choice mattering only at small heterogeneous scales.** All strategies maintained constant 174-byte communication (2.8M× reduction vs centralized). Tested on 1,130-2,709,803 synthetic patients across three scales.

Note: This demonstrates computational feasibility using synthetic Synthea data. Real-world validation requires institutional collaboration.

1. What I Built

A TypeScript tool that: - Computes E-values (sensitivity to unmeasured confounding) at each site - Aggregates using three weighting strategies - Transmits only 174 bytes total (58 bytes per site)

Three strategies tested: 1. **Sample-size (FRI):** $FRI = \sum_k (n_k/N) \cdot E_k$ 2. **Equal-weight:** $(1/K) \sum_k E_k$ 3. **Conservative:** $\min_k \{E_k\}$

E-value formula: $E = RR + \sqrt{RR \times (RR - 1)}$ where RR is risk ratio.

2. Test Results (Synthetic Data)

Table 1: Measured Performance Across Three Scales

Scale	Patients	Sites	FRI (Sample)	Equal- Weight	Conservative	CV
1k	1,130	3	2.015	1.961	1.766	9.7%
100k	235,222	3	2.147	2.147	2.143	0.30%
2.8m	2,709,803	3	2.149	2.149	2.146	0.16%

Key finding: At 1k scale (heterogeneous, CV=9.7%), FRI=2.015 vs min=1.766 (14.1% gap). At 2.8m scale (homogeneous, CV=0.16%), FRI=2.149 vs min=2.146 (0.14% gap). Strategy choice matters only under heterogeneity.

Computational performance (2.8m patients): - Processing time: 10 seconds - Memory: 2-3 GB per site - Communication: 174 bytes (constant)

Communication reduction: - 1k scale: 201 KB $\rightarrow$ 174 bytes ($1,156\times$) - 100k scale: 41.9 MB $\rightarrow$ 174 bytes ($240,805\times$) - 2.8m scale: 482 MB $\rightarrow$ 174 bytes ($2,800,000\times$)

3. What E-values Mean

Interpretation: An E-value of 2.15 means an unmeasured confounder must have risk ratio 2.15 (in both treatment and outcome associations) to nullify the observed effect.

Example (diabetes treatment): - FRI = 2.15 - Known confounders: disease severity (RR~1.8), adherence (RR~1.5) - Since $2.15 > 1.8$ and $2.15 > 1.5$, effect is robust to individual confounders

4. Why This Pattern?

Mathematical property: FRI is bounded by $\min\{E_k\} \leq \text{FRI} \leq \max\{E_k\}$ (convex combination).

Convergence explanation: As sample sizes increase, sampling variation decreases, causing site-level E-values to converge. The coefficient of variation (CV) drops from 9.7% to 0.16%, making all strategies equivalent at large scale.

Sample-size weighting rationale: Follows meta-analysis conventions (DerSimonian & Laird, 1986), giving larger sites proportionally higher weight consistent with their precision contribution.

5. Limitations

Synthetic data only: I used Synthea synthetic healthcare data. Real hospital data may show different heterogeneity patterns. I have no access to real clinical data.

Three sites: Real networks (FDA Sentinel, PCORnet) have 10+ sites. Scaling behavior beyond 3 sites is untested.

Binary outcomes: Continuous outcomes require modified E-value formulas.

Unvalidated thresholds: Proposed thresholds (FRI > 2.0 for clinical guidelines, > 3.0 for FDA approval) lack empirical validation via RCT comparison.

Validation paradox: Synthea has complete confounder knowledge, but E-values quantify robustness to *unmeasured* confounding. Ground truth validation requires comparing observational FRI to RCT outcomes.

Purpose: This is a technical demonstration, not clinical validation. Real-world applicability requires institutional collaboration.

6. Privacy Tradeoff (Design, Not Flaw)

What is protected: Covariate identities. Sites transmit scalar E-values (174 bytes) rather than covariate lists, hiding which adjustment variables were used.

What is not protected: Aggregate effect sizes. E-values are mathematically equivalent to risk ratios via $E = RR + \sqrt{RR \times (RR - 1)}$. This is invertible, so coordinators can infer effect magnitudes.

Why this matters: In meta-analysis, sharing aggregate results is necessary. What federated E-values protect is the “how” (methodological choices), not the “what” (findings). This is a design tradeoff inherent to sensitivity analysis, not a limitation.

Example: Site A may use stigmatizing genetic markers, Site B socioeconomic factors. Federated transmission hides which variables each site chose while sharing robustness conclusions.

7. Reproducibility

```
# Clone repository
git clone https://github.com/watilde/Harmonia
cd Harmonia
```

```
# Install dependencies
npm install
```

```
# Build packages
npm run build
```

```
# Run 1k scale test
npm run test:evalues:1k
```

```
# Run 100k scale test
npm run test:evalues:100k
```

```
# Run 2.8m scale test  
npm run test:evaluates:2.8m
```

Results saved to `research/modules/2-federated-evalues/experiments/results/`

8. What This Demonstrates

Computational feasibility: 2.8m patients in 10 seconds

Communication efficiency: 174 bytes (constant)

Strategy comparison: Three methods tested empirically

Convergence behavior: 14.1% $\rightarrow$ 0.14% gap as heterogeneity drops

Real-world performance: Unknown (synthetic data only)

Clinical validity: No institutional validation

Large networks: 3 sites only

Threshold validation: No RCT comparison

9. Seeking Collaboration

I am an independent OSS engineer without access to real clinical data. This tool is ready for testing on actual multi-site hospital data. If you have OMOP-formatted data and IRB approval, I welcome collaboration to validate these findings in real-world settings.

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References

1. VanderWeele, T. J., & Ding, P. (2017). Sensitivity analysis: introducing the E-value. *Annals of Internal Medicine*, 167(4), 268-274.
 2. DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials*, 7(3), 177-188.
 3. Li, S., et al. (2022). Federated causal inference in heterogeneous observational data. *arXiv:2202.12367*.
 4. Mathur, M. B., et al. (2020). Website and R package for computing E-values. *Epidemiology*, 31(2), e26-e28.
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License: Apache-2.0

Data: Synthea synthetic healthcare data

Ethics: No IRB required (synthetic data only)